S.E.A. College Of Engineering & Technology

32-bit RISC-V Processor Core

Project Review – 1

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Introduction

A processor is the central component of a computing system that executes instructions and performs computational operations.

Modern processors use techniques such as **instruction pipelining** to improve performance and instruction throughput.

RISC-V is an open and modular Instruction Set Architecture (ISA) that provides a flexible foundation for processor design and research.

In this project, we aim to design and implement a **32-bit RISC-V processor core based on the RV32I instruction set**, using **Verilog HDL**.

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Problem Statement

> **To design and implement a functional 32-bit RISC-V processor core based on the RV32I instruction set using Verilog HDL, incorporating a 5-stage instruction pipeline and subsequently verify its functionality through simulation.**

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Objectives

- Understand the RV32I instruction set architecture.
- Design the basic 32-bit processor datapath.
- Implement the processor using Verilog HDL.
- Introduce a 5-stage pipeline: **IF → ID → EX → MEM → WB**
- Implement appropriate hazard handling mechanisms.
- Verify the processor through simulation and test programs.
- Eventually synthesize and evaluate the design.

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Literature Review

Paper	Key Contribution	Relevance to Our Project
"Design and Implementation of 5-Stage Pipelined RISC-V Processor on FPGA"	Presents the design and FPGA implementation of a 5-stage pipelined RISC-V processor.	Provides a direct reference for our **5-stage RV32I processor architecture and FPGA implementation**. (**IEEE Xplore**)
"Design and Analysis of a Multi Clocked Pipelined Processor Based on RISC-V"	Investigates a pipelined processor architecture based on RISC-V and analyzes its implementation.	Helps us understand **pipeline organization and performance considerations** in RISC-V processors. (**IEEE Xplore**)
"Design and Implementation of a Pipelined RV32IMC Processor with Interrupt Support for Large-Scale Wireless Sensor Networks"	Develops a pipelined RISC-V processor with additional instruction-set and interrupt support.	Provides insight into **pipelined RISC-V implementation and future processor extensions**. (**IEEE Xplore**)
"Study of Data Hazard and Control Hazard Resolution Techniques in a Simulated Five Stage Pipelined RISC Processor"	Studies techniques for handling **data and control hazards** in a five-stage pipeline.	Helps guide our implementation of **forwarding, hazard detection and control-hazard handling**. (**IEEE Xplore**)
"Full Open-Source Implementation of an Academic RISC-V on FPGA"	Demonstrates an open-source academic RISC-V implementation on FPGA.	Supports our approach of developing and evaluating a **RISC-V processor using open hardware and FPGA technologies**. (**IEEE Xplore**)

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Existing Systems & Limitations

Existing open-source cores: PicoRV32, Rocket, VexRiscv — offer configurable implementations but often include complexity beyond a basic educational 5-stage pipeline.

Disadvantages

Complex microarchitecture, optional extensions increase difficulty, harder to understand hazard/forwarding logic, not always suitable for lightweight FPGA experiments.

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Proposed System

Modular 5-stage RV32I core with IF, ID, EX, MEM, WB stages. Key components :

- **Register File**

  32 × 32-bit registers

- **Immediate Gen**

  Sign-extend/immediate generation

- **Hazard Unit**

  Load-use stalling and control hazards

- **ALU**

  32-bit ALU

- **Forwarding Unit**

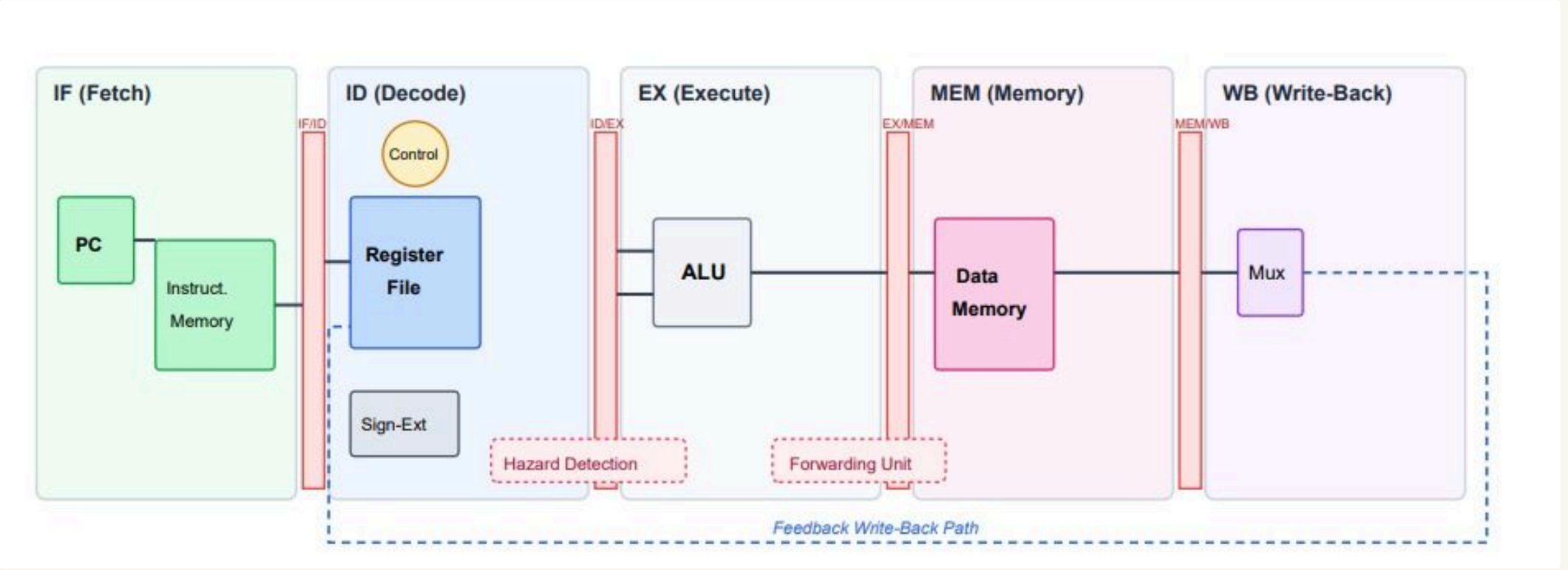
  Data dependency resolution

- **CSR & Trap**

  Basic CSR and exception handling

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Block Diagram



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Methodology

Development steps:

1. Define RV32I instruction support
2. Create pipeline data path and control logic
3. Implement inter-stage registers
4. Add forwarding and stalling mechanisms
5. Implement memory interface and CSR handling
6. Develop Verilog testbench
7. Run simulation and compliance verification
8. Synthesize for FPGA and analyze timing

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Hardware & Software Requirements

Hardware

- FPGA board (Xilinx / Intel / Lattice)
- 4 GB RAM minimum
- Dual-core processor or higher processor

Software

- Verilog HDL
- Verilator / ModelSim
- Vivado
- RISC-V GCC Toolchain
- Spike Simulator

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Advantages & Applications

Advantages

Simple, modular, educational, high instruction throughput, low resource usage, synthesis-friendly RTL, supports RV32I software, extensible for future ISA enhancements.

Applications

Embedded systems, FPGA learning platforms, computer architecture labs, custom SoC development, RISC-V research and prototyping.

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Expected Outcome & Conclusion

Expected: A functionally correct 5-stage RV32I pipelined core that passes directed and ISA compliance tests, approaches one instruction per cycle under normal operation, and runs successfully on FPGA with practical timing.

Conclusion: The core balances simplicity and performance via forwarding, hazard detection, branch handling, and minimal privileged support—suitable for academic learning, FPGA use, and future extension.

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References

- RISC-V Unprivileged ISA Specification, Version 2026.
- RISC-V Privileged Architecture Specification, Version 2026.
- PicoRV32 Open-Source Core Documentation & Verification Suite.
- VexRiscv Open-Source Core Microarchitecture Guidelines.
- Verilator Engine and AMD Xilinx Vivado Reference Tool Manuals.

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THANK YOU

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